

# SUMIT SINGH

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## PROFESSIONAL SUMMARY

AI & Data Science undergrad (CGPA 8.6/10) with 3 internships and shipped products across the ML stack: NLP pipelines, LLM agents and RAG systems, document-extraction pipelines, and the backend infrastructure that serves them at production traffic. 600+ DSA problems, 4x hackathon winner, GDSC Technical Head. Seeking full-time AI / ML Engineer roles.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, SQL, C++ (DSA)

**ML / AI:** NLP pipelines (entity & location extraction), LLM agents & structured output, RAG retrieval, prompt engineering, generative-AI integration, feature engineering, model evaluation

**Libraries & Tools:** PyTorch, Hugging Face Transformers, NumPy, pandas, scikit-learn, Tesseract OCR, Playwright, Git, Jupyter/Colab

**Systems & Backend:** Node.js, Express.js, Next.js, REST APIs, MongoDB, Redis, Caching, Rate Limiting, Webhooks, React.js

## AI / ML PROJECTS

### JanSetu Health System · [Live](#) · [GitHub](#)

*Next.js, MongoDB, Gemini, NLP, RAG, OpenStreetMap APIs*

- Built a rural disease-surveillance platform: ASHA health workers speak field reports, a custom NLP parser extracts disease entities and case counts, and a live map updates with risk zones, trends, and outbreak alerts — replacing days of manual HQ reporting across 8 MongoDB collections, 26 API routes, and 4 user roles.
- Designed a correlation engine cross-checking two independent, imperfect data streams — worker-reported cases vs. actual medicine movement at local shops — to flag outbreaks and under-reporting that neither source catches alone; divergence between streams auto-generates "go survey this village" alerts.
- Shipped an LLM health chatbot (Gemini + RAG retrieval) that guides residents to OTC medicines and reserves them at nearby pharmacies via 4-digit pickup codes; prescription photos are read by Gemini vision with structured JSON output.
- Enforced "rules decide, the model only speaks": every safety-critical decision (emergency detection, OTC/RX gating, risk scoring) is deterministic code — remove the Gemini key and the whole app still works with degraded language quality and lexical-only search.

### AI Document Assist · [Live](#) · [GitHub](#)

*React, Node.js, Express, MongoDB, Python, Gemini, Tesseract OCR, Chrome MV3*

- Built a hybrid document-extraction pipeline — Gemini vision first, escalating to Tesseract OCR + 25 regex extractors, then a Gemini text fallback over the OCR output — parsing 6 government document types into 31 canonical fields, with min-of-token confidence scoring so one wrong character invalidates a structured ID.
- Scored scheme eligibility with a 7-signal weighted engine (occupation, eligibility rules, state, demographics); Gemini re-ranks the top matches, with the deterministic ranking retained as fallback. A Python crawler (with a circuit-breaker rate limiter) syncs the MyScheme catalogue.
- Built a Chrome MV3 extension that replays AI-generated field mappings to auto-fill multi-step government application forms end-to-end — final submission always user-controlled — across 23 JWT-protected Express routes with Cloudinary document storage.

## EXPERIENCE

### Software Development Engineer | SwapSo

May 2026 – Present · Remote

- Architected a provider-abstracted gift-card rewards integration for a crypto wallet: event-driven webhook processing with HMAC-SHA256 verification, idempotent 1.25% USDC cashback engine, AES-256-GCM encryption, and 400+ brand catalogue sync.
- Built OTP authentication backend (Node.js, TypeScript, MongoDB, Redis) with TTL-based expiry, rate limiting, CAPTCHA, and retry/fallback logic — cutting DB load and improving verification latency under traffic spikes.

### Software Developer (Consultant) | The Sibling Corp

Sep 2025 – Dec 2025 · Remote

- Designed and shipped Quick Basket — farmer-to-consumer marketplace (Node.js, Express.js, MongoDB, React Native) launched on Android and iOS; architected the REST APIs, data models, and schemas for end-to-end order management.

### Frontend Developer Intern | Growigh

Jul 2025 – Sep 2025 · Remote

- Built a responsive React.js landing page against backend REST APIs; raised Lighthouse score from 64 to 91/100 via code splitting and asset optimization — 40% faster load time, 25% higher avg. session duration.

## EDUCATION

**B.Tech — Artificial Intelligence & Data Science**, University of Mumbai (TCET) · CGPA 8.6/10

Aug 2023 – Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Linear Algebra & Probability, DSA, DBMS.

HSC (Class XII), Maharashtra State Board · 2023 · 70.17% | SSC (Class X), Maharashtra State Board · 2021 · 79.40%

## ACHIEVEMENTS & LEADERSHIP

- 4x Hackathon winner · SIH Institute-Level Finalist · GSSoC 2024 Open Source Contributor · 600+ DSA problems solved.
- **Technical Head — GDSC, TCET Mumbai:** delivered technical workshops for 200+ student developers. **Problem Setter — Shastra Coding Club:** authored competitive programming problems for inter-college contests.